C-GR.2.2 · Weekly check · 5 questions

MA.7.GR.2.2 · Real-world cylinder surface area · Unit 12 Surface Area and Volume · Given Thursday, on the week's benchmark

Form title C-GR.2.2 Check · Real-world cylinder surface area
Description Learning goal: I can solve problems involving surface area of cylinders. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

Short-answer validation (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

#	Question image	File · type	Options to add	Answer key	Pts
1	Surface area of a right circular cylinder: $SA = 2\pi r^2 + 2\pi rh$ A closed cylindrical water tank is made from sheet metal. The tank has a radius of 9 feet and a height of 17 feet. What is the total area of sheet metal needed for the curved side, the top and the bottom? Write your answer in terms of π. Type only the number in front of π. Example: for 7π, type 7.	**Q1.png** Short answer	none (typed answer)	**468** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
2	Surface area of a right circular cylinder: $SA = 2\pi r^2 + 2\pi rh$ A cylindrical container is made of plastic. It has a radius of 3 inches and a height of 14 inches, and it has a bottom but no top. What is the area of the plastic needed to make the container? Write your answer in terms of π. Type only the number in front of π. Example: for 7π, type 7.	**Q2.png** Short answer	none (typed answer)	**93** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
3	Surface area of a right circular cylinder: $SA = 2\pi r^2 + 2\pi rh$ A cylindrical gift tin with a lid has a radius of 3 inches and a height of 4 inches. The whole tin, including the lid and the bottom, is covered with decorative paper. What is the area the paper covers, in square inches? A. 42π B. 30π C. 33π D. 24π	**Q3.png** Multiple choice	A B C D	**A**	1
4	Surface area of a right circular cylinder: $SA = 2\pi r^2 + 2\pi rh$ A large cylindrical storage container has a radius of 5 inches and a height of 11 inches. A label wraps all the way around its curved side, with no overlap; the top and the bottom are not covered. What is the area of the label? Round your answer to the nearest hundredth. Use the π key on your calculator. Type a number only. No units, $ or %.	**Q4.png** Short answer	none (typed answer)	**345.58 · 345.4 · 345.40** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
5	Surface area of a right circular cylinder: $SA = 2\pi r^2 + 2\pi rh$ A cylindrical water tank stands on the ground. It has a radius of 7 feet and a height of 17 feet. The curved side and the top will be painted, but not the bottom. One gallon of paint covers 400 square feet, and paint is sold only in whole gallons. How many gallons of paint must be bought? Use the π key on your calculator. Type a number only. No units, $ or %.	**Q5.png** Rigor Short answer	none (typed answer)	**3 · 3.0 · 3.00** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
6	LEARNING GOAL I can solve problems involving surface area of cylinders. 4 I can use surface area to solve a real problem (paint, cost, material), including when only some faces count. 3 I can solve a real-world problem with the surface area of a cylinder. 2 I can find the area of a circle and a rectangle. 1 I need help to start this.	**Level scale.png** Multiple choice (the template's last question)	Level 1 Level 2 Level 3 Level 4	no key	0

Answers with worked solutions are in the Answers folder next to the question images (same file names).